

Name:

Key

9. a) For each of the following two matrices over  $\mathbb{C}$ , list its eigenvalues and give a basis for each of its eigenspaces:

$$A = \begin{pmatrix} 5 & 1 \\ -3 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 5-\lambda & 1 \\ -3 & 1-\lambda \end{vmatrix}$$

$$= (5-\lambda)(1-\lambda) + 3$$

$$= \lambda^2 - 6\lambda + 5 + 3 = \lambda^2 - 6\lambda + 8$$

$$\text{So } \lambda^2 - 6\lambda + 8 = 0 \Rightarrow (\lambda - 4)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 4 \text{ or } 2$$

$$\text{null}(A - 4I): \begin{pmatrix} 1 & 1 \\ -3 & -3 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\text{so the } 4\text{-eigenspace} = \text{span} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\text{null}(A - 2I): \begin{pmatrix} 3 & 1 \\ -3 & -1 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} 3 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\text{so the } 2\text{-eigenspace} = \text{span} \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

since  $B$  is diagonal,  
we have that  $\lambda = 2, 9, 9$   
are the eigenvalues  
and that a basis for  
the two eigenspaces are:

$$\lambda = 2: \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\lambda = 9: \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

b) For each of the matrices  $A$  and  $B$  above, give a diagonalization of it ( $C = PDP^{-1}$ ); or explain why this is not possible.

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -3 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -3 \end{pmatrix}^{-1}$$

eigenvectors

corresponding  
eigenvalues

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1}$$

Remember: every  
diagonal matrix is  
diagonalizable!

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9. a) For each of the following two matrices over  $\mathbb{C}$ , list its eigenvalues and give a basis for each of its eigenspaces:

$$A = \begin{pmatrix} 10 & 3 \\ 3 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 10 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 10-\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix}$$

$$= (10-\lambda)(2-\lambda) - 9$$

$$= \lambda^2 - 12\lambda + 20 - 9$$

$$= \lambda^2 - 12\lambda + 11$$

If  $\lambda^2 - 12\lambda + 11 = 0$  then

$$(\lambda - 11)(\lambda - 1) = 0 \Rightarrow \lambda = 11 \text{ or } 1.$$

$$\text{null}(A - 11I):$$

$$\begin{pmatrix} -1 & 3 \\ 3 & -9 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} -1 & 3 \\ 0 & 0 \end{pmatrix} \Rightarrow 11\text{-eigenspace} = \text{span} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\text{null}(A - 1I):$$

$$\begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} 3 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow 1\text{-eigenspace} = \text{span} \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

Since  $B$  is upper triangular, we have  
 $\lambda = 3, 3, 10.$

$$\text{null}(B - 3I): \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 10 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 10 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$3\text{-eigenspace} = \text{span} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ REF, 2 pivots}$$

$$\text{null}(B - 10I): \begin{pmatrix} -7 & 1 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ REF, 2 pivots}$$

$$10\text{-eigenspace} = \text{span} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

b) For each of the matrices  $A$  and  $B$  above, give a diagonalization of it ( $C = PDP^{-1}$ ); or explain why this is not possible.

$$A = \begin{pmatrix} 3 & 1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 11 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 1 & -3 \end{pmatrix}^{-1}$$

eigenvectors

corresponding eigenvalues

$B$  is not diagonalizable.

For  $B$ , for  $\lambda = 3$ , algebraic multiplicity  $>$  geometric multiplicity  
 multiplicity as root of characteristic polynomial  $\rightarrow 2 > 1 \leftarrow$  dimension of eigenspace